Econometrics I

Lecture 6: Instrumental Variables I — endogeneity, simultaneity, 2SLS

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- Basic Idea of Instrumental Variable (IV):
  - What if we have a variable that is correlated with X *but not with* Y
  - Then any changes in Y caused by that variable will reflect causal changes by X
- Equation of interest:

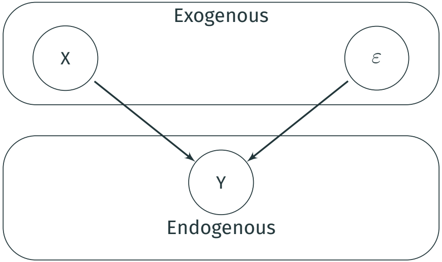
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- IVs break X into two pieces that are themselves uncorrelated:

$$X_i = \gamma_0 + \gamma_1 Z_i + \eta_i$$

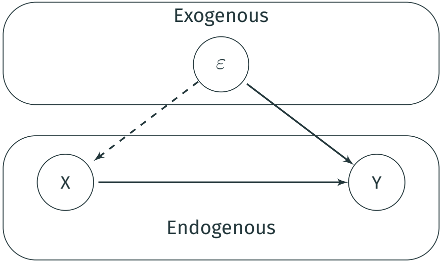
  - A piece that is not correlated with ε ($Cov(Z, \varepsilon) = 0$)
  - A piece that is correlated with ε ($Cov(\eta, \varepsilon) \neq 0$) – source of the endogeneity problem
  - Finally, $Cov(Z, \eta) = 0$
- Z_i is an **instrumental variable**

A “Good” Regression:



- **Exogenous Variables:** Variables in the data that do not cause each other
  - ε is always exogenous, so exogenous also just means variables not correlated with ε
- **Endogenous Variables:** Variables that are determined by exogenous variables in the model
  - ε is always in Y so Y is always endogenous

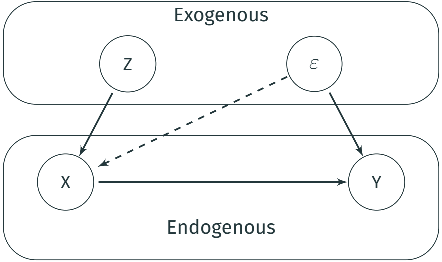
Selection/OVB:



- In this picture X is **endogenous** because ε now causes X as well
- What if there is another exogenous variable that *does not* directly cause Y?

Omitted Variable Bias with Pictures

Instrumental Variable:



- Z causes Y *only indirectly* through X
- We can estimate “causal effect” of Z on Y and this MUST be the causal effect of Z on X and the causal effect of X on Y
  - Mathematically we need to split effect of on Z on Y into effect of Z on X and X on Y
  - Related concept in statistics: Path Analysis

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- We call a variable Z a a valid **instrumental variable** if the following two conditions hold:

1. **Relevance:** $Cov(X, Z) \neq 0$
   - An arrow from Z to X in the pictures
2. **Exogeneity:** $Cov(\varepsilon, Z) = 0$
   - No arrow from Z to Y or Z to ε in the picture

Key Assumption #1 of Instrumental Variables

- **Relevance:** $Cov(X, Z) \neq 0$
- This assumption just means that X and Z are correlated
- We observe both X and Z, so can easily test this assumption by regressing:

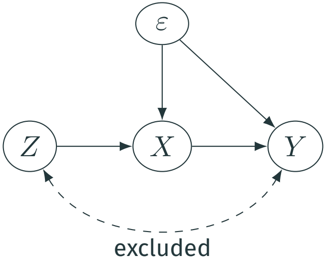
$$X_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$$

- If $\beta_1 \neq 0$ in regression, we say instrument is relevant

Key Assumptions #2 of Instrumental Variables

- **Exogeneity (or validity):** $Cov(\varepsilon, Z) = 0$
- This assumption means that Z and ε cannot be correlated
- We do not observe ε, so we **cannot** test this assumption
- In general, we need to "defend" this assumption by telling a story about why Z and ε are unlikely to be correlated
- Much like parallel trends or strict exogeneity, these crucial identifying assumptions cannot be tested on their own. However, we can test them against each other. That is, the best we can do is ask the question "given this exogeneity assumption, does this other exogeneity assumption seem true?" But we can never test an exogeneity assumption in a self-contained way.

- Recall the absence of direct path between $Z \rightarrow Y$. This is the exclusion restriction.
- $Z \rightarrow X$ is the relevance condition.



Idea: There is a path from $Z \rightarrow Y$ but only through X (this is the key).

Best Defense of Exogeneity IV Assumption: Randomized Experiment

- Back to the class size example:

$$score_i = \beta_0 + \beta_1 \text{CS}_i + \varepsilon_i$$

- where:
  - $score_i$: Test score of student i
  - CS_i: Class size of student i

- Suppose that we use a coin flip that sends kids that get a “head” to a small class and kids getting a “tails” to big class
  - This is our randomized experiment!

- Let’s call the coin flip our instrument Z (where $Z_i = 1$ if heads, $Z_i = 0$ if tails)

Best Defense of Exogeneity IV Assumption: Randomized Experiment

$$score_i = \beta_0 + \beta_1 \text{CS}_i + \varepsilon_i$$

- Is Z (our coin flip) a good instrument?
- **Relevance:** $Cov(CS, Z) \neq 0$? Yes, if $Z_i = 1$ kid gets small class, if $Z_i = 0$ kid gets big class
    - So the regression $CS_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$ will estimate that $\beta_1 < 0$
- **Exogeneity:** $Cov(\varepsilon, Z) \neq 0$? Untestable – so need to tell a story
    - Story: Exogeneity holds because coin flip is random and does not depend on any student or parent characteristics that would affect test scores. Therefore, there is nothing related to Z (besides X) that is also related to test scores, so ε and Z must be uncorrelated.

      This is a good story! It's the "magic" of randomization.

Randomized Experiment as an IV

- So a randomized experiment can be treated as an IV, but there is a big difference in how we evaluate the assumption for an experimental or non-experimental setting
- When running an experiment, the exogeneity assumption is justified if the randomization was implemented in way that was not correlated with anything else that could influence outcomes. Attrition and manipulation can undermine exogeneity in an experimental context if they create a correlation between treatment status and other factors that might influence outcomes. Thus, exogeneity amounts to an assumption about how the randomization was implemented.
- In contrast, the IV exogeneity assumption in a non-experimental context is a broad and vague statement about how the world works; it's much more difficult to interpret and evaluate. The same is true of strict exogeneity or parallel trends.

IV Estimation: Example with a binary $Z \in \{0,1\}$

Let's begin with the simplest 0/1 model (this time for Z):

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- X is endogenous but we have a 0/1 instrument, Z
  - Because Z is 0/1 we can focus on group means.
- What assumptions do we need to make?
  - $Cov(X, Z) \neq 0$
  - $Cov(Z, \varepsilon) = 0$
- The intuition from above: solve for the "indirect" effect of Z on Y

The Relationship Between Y on Z

What is the relationship between Y and Z, taking expectations?

$$\begin{aligned}
\mathbb{E}(Y|Z) =& E\left(\beta_0 + \beta_1 X + \varepsilon | Z\right) \\
=& \beta_0 + \beta_1 \mathbb{E}(X|Z) + \underbrace{\mathbb{E}(\varepsilon|Z)}_{=0} \\
=& \beta_0 + \beta_1 \mathbb{E}(X|Z)
\end{aligned}$$

Since Z is 0/1...

$$\begin{aligned}
\mathbb{E}(Y|Z=1) =& \beta_0 + \beta_1 \mathbb{E}(X|Z=1) \\
\mathbb{E}(Y|Z=0) =& \beta_0 + \beta_1 \mathbb{E}(X|Z=0)
\end{aligned}$$

Rearranging...

$$\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0) = \beta_1 \times \left(\mathbb{E}(X|Z=1) - \mathbb{E}(X|Z=0)\right)$$

From the previous slide we have:

$$\beta_1 = \frac{\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0)}{\mathbb{E}(X|Z=1) - \mathbb{E}(X|Z=0)}$$

From the LLN we can construct the following estimator:

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- The subscript refers to $Z = 1$ or $Z = 0$
- We call this the IV estimator

Estimator:

$$\hat{\beta}_1^{IV} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}$$

- For a binary X we look at the change in Y over the change in X:

$$\hat{\beta}_1^{OLS} = \frac{\bar{Y}_{X=1} - \bar{Y}_{X=0}}{\bar{X}_{X=1} - \bar{X}_{X=0}} = \Delta\bar{Y}$$

  - For a $0/1$ X, $\Delta\bar{X} = 1$
  - Basically looking at how much Y changes for a change in X
- Intuition for a 0/1 Z
  - The effect of X on Y is still the change in Y over a change in X
  - Use the change in Y *given* Z to shut down the effects of ε
  - Use the change in X *given* Z to get the right scaling

The IV Regression with a Continuous Z

Same model as before:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Now assume that Z is continuous
- How to capture the indirect effect of Z?
  - Intuition from OLS: use the covariance between Z and Y as a measure of the relationship:

$$\begin{aligned} Cov(Y,Z) =& Cov(\beta_0 + \beta_1 X + \varepsilon, Z) \\ =& \beta_1 Cov(X,Z) \end{aligned}$$

  - Rearranging:

$$\beta_1 = \frac{Cov(Y,Z)}{Cov(X,Z)}$$

- **Instrumental Variables Estimator:** $\hat{\beta}_1^{IV} = s_{YZ}/s_{XZ}$

- Suppose that we are interested in investigating the effect of studying on grades:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- where
  - Y_i: is GPA
  - X_i: is study time (hours per day)
- We expect that our OLS estimator $\widehat{\beta}_1$ will be severely biased here (why?)

- To get rid of OVB, we shall use an IV
- One possible IV: whether your roommate has a N64 (or playstation/whatever video game console kids use today)
- Important feature: Roommates are randomly assigned in college
  - At least at Berea College (Kentucky) where this example comes from (see Stinebrickner and Stinebrickner (2008))

- First thing for any instrument is to think of validity. Two conditions:
    - Instrument is $N64$, which is indicator variable that your roommate has N64

1. Relevance: $Cov(Z, X) \neq 0$; here $Cov(N64, study) \neq 0$
    - Seems likely to hold as everyone prefers playing Mario Kart to studying
    - Testable
2. Exogeneity: $Cov(Z, \varepsilon) = 0$; here $Cov(N64, \varepsilon) = 0$
    - Untestable
    - Is it plausible?

Thinking About Exogeneity

- Exogeneity: $Cov(Z, \varepsilon) = 0$
- "Storytime" should talk about how two things hold
  1. People select into Z in some manner that is likely to be correlated with Y
     - Concern: People that pick roommates that are "fun" and have N64s are likely people who do not care too much about grades
  2. Z only affects Y through X
     - Concern: Having a roommate with a video game affects your GPA through other means than affecting your study hours
- Either story "invalidates" the instrument (but in different ways)

1. Selection story: there is an omitted variable related to both Z and Y
    - Example: omitted variable is effort because students that really put in a lot of effort make sure they do not get a roommate with N64

2. Other channel story:
    - Possible Story 1: Mario Kart helps me grasp physics, so I ace my physics exam
        - so Z directly affects Y *independent* of X
    - Possible Story 2: Other people hang out in our room due to our N64, making it really loud and so I cannot study effectively
        - so Z affects Y through *another* X

- Conceptually, there are two aspects to the exogeneity assumption: **randomization** and **exclusion**. Be careful: even when an IV is clearly random, it might not be excludable.

IV Math

IV Directly into Model (aka "the reduced form")

- Suppose that our IV assumptions hold
- Then we can just directly replace X with our IV in our model:

$$Y_i = \pi_0 + \pi_1 Z_i + \varepsilon_i$$

- where
  - Y_i: is GPA
  - Z_i: is whether roommate has N64
- **Interpretation of $\widehat{\beta}_1$:** Having a roommate with a N64 **causes** a $\widehat{\beta}_1$ decrease in GPA
  - Causal effect because validity of instrument, $cov(Z, \varepsilon) = 0$, implies OLS is consistent for above equation. Of course we could question the instrument's validity.
- But we want the effect of *studying* on GPA!

A popular IV estimator is Two Stage Least Squares (2SLS)

- This effectively runs regression $Y_i = \beta_0 + \beta_1 Z_i + \varepsilon_i$, but *scales* β_1 by how much Z affects X
  - If Z affects X at one-to-one rate, β_1 in above regression will capture effect of X
  - If Z affects X only a little, we need to scale β_1 up a lot to say what effect of X on Y is
- We therefore estimate in two steps:
  1. Regress X on Z to capture effect of Z on X
  2. Regress fitted values of step 1 to capture effect of X on Y
- Under IV assumptions, this gives us **causal** effect of X on Y

- Step 1: regress

$$X_i = \pi_0 + \pi_1 Z_i + \eta_i$$

- Step 2: take your predicted values from Step 1, $\hat{X}_i$, and regress

$$Y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$$

- Under IV assumptions, step 1 finds the "good part" of X (that is not related to ε), and then step 2 takes that "good part" of X as a regression to find the causal relationship between X and Y

- Why is $\hat{\beta}_1^{2SLS} = \frac{Cov(Y,Z)}{Cov(X,Z)}$?
- True model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Take covariance of both sides with respect to Z:

$$Cov(Y_i, Z_i) = Cov(\beta_0 + \beta_1 X_i + \varepsilon_i, Z_i)$$
$$Cov(Y_i, Z_i) = Cov(\beta_0, Z_i) + Cov(\beta_1 X_i, Z_i) + Cov(\varepsilon_i, Z_i)$$
$$Cov(Y_i, Z_i) = 0 + \beta_1 Cov(X_i, Z_i) + 0 \;\; \textit{since } Cov(\varepsilon_i, Z_i) \textit{ by assumption}$$
$$\implies \beta_1 = \frac{Cov(Y,Z)}{Cov(X,Z)}$$

Single Variable IV in R: Manually

- Data:
  - Y_i: is GPA
  - X_i: is study time (hours per day)
  - Z_i: is indicator for roomate having N64
- Running 2SLS by doing both steps manually:
- Step 1: Regress $X_i = \pi_0 + \pi_1 Z_i + v_i$ and get predicted study time, $\hat{X}_i$
  - `Data$xHat = predict(feols(X~Z, data=Data))`
- Step 2: Regress $y_i = \gamma_0 + \gamma_1 \hat{X}_i + \varepsilon_i$
  - `m3 = feols(Y~xHat, data=Data)`
  - `summary(m3, vcov= "HC1"))`
    - Note: Standard errors will be wrong!
  - `predict` gives the fitted values of X from the first stage
  - Then `feols` can be used (again) to get the second stage
  - Problem: the standard errors (even robust) will be wrong

R can run 2SLS all at once:

- `m4 = feols(Y ~ W | X ~ Z, data=Data)`
- First formula: Regress Y on exogenous controls W (and endogenous X)
- Second formula: Write first-stage regression for endogenous X (included in second stage automatically)
- `feols` will also adjust standard errors

This is basically why we use `feols` whenever we can.

IV Setup with Matrix Notation

- Consider a multiple regression equation

$$y = \beta X + \varepsilon$$

The IV assumptions are now that

  - $Cov\left(z_i, \varepsilon_i\right) = 0$
  - $Cov\left(z_i, x_i\right)$ has *full rank*, i.e., rank K, where K is the number of regressors in **x**

- We can have multiple regressors as well as instruments. Some regressors can also be instruments; if $z_i = x_i$ then we can just use OLS.
- Regressors that are instruments are **exogenous regressors**. Regressors that are not instruments are **endogenous regressors**.
- For the rank assumption, we need at least one instrument for each endogenous regressor.

- The 2SLS formula is now

$$b_{2SLS} = \left(\hat{X}'\hat{X}\right)^{-1}\hat{X}'y$$

$$\text{with } \hat{X} = Z\left(Z'Z\right)^{-1}Z'X$$

- With homoscedasticity, the asymptotic variance of the estimator is

$$Var\left(b_{2SLS}\right) = \frac{s^2}{n}\left[Z'X\left(X'X\right)^{-1}X'Z\right]^{-1}$$

- Note: this (asymptotically correct) estimate of the variance is not equal to the (incorrect) formula we would get from naïvely applying OLS formulas to the second-stage regression.

Doing General IV Case in R

- Before the |: $Y \sim Controls$ (endogenous X is automatically included)
- After the |: $X \sim Z$ (excluded instruments only, the exogenous controls are included automatically)
- Syntax differs for different packages $\rightarrow$ Be careful!

```r
library(fixest)
library(tidyverse)
library(ddml)
library(car)

babydata <- data.frame(AE98) %>%
  mutate(BoyBoy = 1* (boy1st ==1 & boy2nd==1),
         GirlGirl = 1* (boy1st ==0 & boy2nd==0))

m5 = feols(workedm | morekids + boy1st + boy2nd + black + hispan +
      othrace + agem1 + agefstm | samesex ~ boy1st + boy2nd,
     data=babyData)
```

For the single-variable case, we can show that

$$Var(\hat{\beta}_{2SLS}) = \frac{1}{N}\frac{\sigma_\varepsilon^2}{\sigma_X^2 R_{XZ}^2}\mathbf{X}'\mathbf{e}_{OLS} = \mathbf{0}$$

OLS variance under homoscedasticity: $Var(\hat{\beta}^{OLS}) = \frac{1}{N}\frac{\sigma_\varepsilon^2}{\sigma_X^2}$

- Since $R_{XZ}^2 \leq 1$, IV variance is larger
- The higher first stage R_{XZ}^2, the lower the variance. If X and Z are collinear, variance is equivalent to OLS.
- If $R_{XZ}^2 \approx 0$ then variance blows up!

Unbiasedness “Proof”

$$\mathbb{E}[\hat{\beta}_1^{2SLS}] = \mathbb{E}\left[\frac{\sum_{i=1}^{N}(z_i - \bar{z})(y_i - \bar{y})}{\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})}\right]$$

$$= \beta_1 + \mathbb{E}\left[\frac{\sum_{i=1}^{N}(z_i - \bar{z})\varepsilon_i}{\sum_{i=1}^{N}(x_i - \bar{x})(z_i - \bar{z})}\right] = \beta_1 + 0$$

Simultaneous Equations

New Model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$
$$X_i = \alpha_0 + \alpha_1 Y_i + \nu_i$$

- In many economic environments, variables both determine each other "in equilibrium" or are determined together "jointly"
- A model where X and Y "cause" each other is called a **simultaneous equations model**
- Examples:
  - Angrist & Evans is an example of this: labor supply and fertility decisions are determined jointly, there is no one-way channel
  - Supply and demand is the canonical example: supply equations and demand equations are of interest but in equilibrium only observe $Q_S = Q_D$
  - Also common in macroeconomic contexts: interest rates, unemployment rates, etc.

Supply and Demand For Coffee, everything is linear

$$
\begin{aligned}
Q_t^d &= \alpha_0 + \alpha_1 P_t + U_t \\
Q_t^s &= \beta_0 + \beta_1 P_t + V_t \\
Q_t^d &= Q_t^s
\end{aligned}
$$

Solving for P_t, Q_t:

$$
\begin{aligned}
P_t &= \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{V_t - U_t}{\alpha_1 - \beta_1} \\
Q_t &= \frac{\alpha_1 \beta_0 - \alpha_0 \beta_1}{\alpha_1 - \beta_1} + \frac{\alpha_1 V_t - \beta_1 U_t}{\alpha_1 - \beta_1}
\end{aligned}
$$

Price is a function of both error terms, and we can't use a clever substitution to cancel things out.

To make things really obvious:

$$Cov(P_t, U_t) = -\frac{Var(U_t)}{\alpha_1 - \beta_1}$$
$$Cov(P_t, V_t) = \frac{Var(V_t)}{\alpha_1 - \beta_1}$$

- When demand slopes down ($\alpha_1 < 0$) and supply slopes up $(\beta_1 > 0)$ then price is positively correlated with demand shifter U_t and negatively correlated with supply shifter V_t.

$$\begin{aligned} Cov(P_t, Q_t^d) &= \alpha_1 Var P_t + Cov(P_t, U_t) \\ Cov(P_t, Q_t^s) &= \beta_1 Var P_t + Cov(P_t, V_t) \end{aligned}$$

- Bias in OLS estimate (Demand) $Bias(\alpha_1) = \frac{Cov(P_t, U_t)}{Var P_t}$.
- Bias in OLS estimate (Supply) $Bias(\beta_1) = \frac{Cov(P_t, V_t)}{Var P_t}$.
- We can actually write both this way when $Cov(U_t, V_t) = 0$:

$$\text{OLS Estimate} = \frac{\alpha_1 Var(V_t) + \beta_1 Var(U_t)}{Var(V_t) + Var(U_t)}$$

- More variation in supply V_t $\rightarrow$ better estimate of demand.
- More variation in demand U_t $\rightarrow$ better estimate of supply.
- Led Working to say the statistical demand function (OLS) is not informative about the economic demand function (or supply function).

IV Can Solve Simultaneity Bias

Why cover simultaneity bias *now*?

New Model:

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 X_i + \varepsilon_i \\ X_i =& \alpha_0 + \alpha_1 Z_i + \alpha_2 Y_i + \nu_i \end{aligned}$$

- If we have data on Z, we can potentially use Z as an instrument!
- What conditions would Z have to fulfill?
  1. Instrument Relevance: $\alpha_1 > 0$
  2. Instrument Exogeneity: $Cov(Z_i, \varepsilon_i) = 0$

Model:

$$
\begin{aligned}
Hours_i =& \beta_0 + \beta_1 Children_i + \varepsilon_i \\
Children_i =& \alpha_0 + \alpha_1 Hours_i + \alpha_2 SameSex_i + \nu_i
\end{aligned}
$$

- $SameSex$: indicator for whether first two children have same sex
- Hours and children are determined jointly
- Also $Cov(\varepsilon, \nu)$ is unlikely to be 0
  - Education can matter for both number of children and hours worked (so education is part of ε and ν)
  - Being married may matter for both
  - Quality of non-financial benefits (e.g., maternity leave)
- So formally, the instrument $SameSex$ is supposed to help solve simultaneity

TABLE 8—OLS AND 2SLS ESTIMATES OF LABOR-SUPPLY MODELS USING 1990 CENSUS DATA

	All women			Married women			Husbands of married women		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Estimation method	OLS	2SLS	2SLS	OLS	2SLS	2SLS	OLS	2SLS	2SLS
Instrument for *More than 2 children*	—	*Same sex*	*Two boys, Two girls*	—	*Same sex*	*Two boys, Two girls*	—	*Same sex*	*Two boys, Two girls*
Dependent variable:									
Worked for pay	−0.155 (0.002)	−0.092 (0.024)	−0.092 (0.024) [0.743]	−0.147 (0.002)	−0.104 (0.024)	−0.104 (0.024) [0.576]	−0.102 (0.001)	0.017 (0.009)	0.017 (0.009) [0.989]

Example: Porter (1983)

- The Joint Executive Committee's (19th Century railroad cartel) main business was transporting grain from Midwest to ports on East Coast.
- Porter (1983) estimates the following simultaneous system:

$$\begin{aligned} \text{demand:} \quad \log Q_t &= \alpha_0 + \alpha_1 \log P_t + \alpha_2 L_t + U_{1t} \\ \text{supply:} \quad \log P_t &= \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t} \end{aligned}$$

where

- Q_t is the quantity (tonnage of grain) transported by the JEC
- P_t is the price per ton of grain transported
- L_t is a dummy for whether the Great Lakes were open to navigation
- S_t is a vector of time dummies
- I_t is an indicator for whether the JEC was colluding

Example: Porter (1983)

$$
\begin{aligned}
\text{demand: } \log Q_t &= \alpha_0 + \alpha_1 \log P_t + \alpha_2 L_t + U_{1t} \\
\text{supply: } \log P_t &= \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t}
\end{aligned}
$$

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\end{aligned}
$$

Example: Porter (1983)

TABLE 3 Estimation Results*

Variable	Two Stage Least Squares (Employing *PO*) Demand	Supply
C	9.169 (.184)	−3.944 (1.760)
LAKES	−.437 (.120)	
GR	−.742 (.121)	
DM1		−.201 (.055)
DM2		−.172 (.080)
DM3		−.322 (.064)
DM4		−.208 (.170)
PO/PN		.382

- How much does the biofuels mandate increase the price of grains? It depends on supply and demand.
- Roberts and Schlenker (2013) estimate the following supply and demand equations:

$$\begin{aligned} \text{demand:} \quad \log Q_{dt} &= \alpha_0 + \alpha_1 \log P_{dt} + \varepsilon_{dt} \\ \text{supply:} \quad \log Q_{st} &= \beta_0 + \beta_1 \log P_{st} + \beta_2 \omega_t + \varepsilon_{st} \end{aligned}$$

  - Q_{st} is global grain supply (calories)
  - Q_{dt} is global grain consumption ($Q_{st} \neq Q_{dt}$ because of storage)
  - The prices are measured at different times (planting vs harvest time)
  - ω_t is a yield shock (due to weather), used as an instrument for demand
  - ω_{t-1} is used as an instrument for supply (through storage, past production matters for current prices)

Example: Roberts and Schlenker (2013)

TABLE 1—SUPPLY AND DEMAND ELASTICITY (*FAO data*)

	Instrumental variables			Three-stage least squares		
	(1a)	(1b)	(1c)	(2a)	(2b)	(2c)
Panel A. Supply equation						
Supply elast. β_s	0.102***	0.096***	0.087***	0.116***	0.112***	0.097***
	(0.025)	(0.025)	(0.020)	(0.019)	(0.020)	(0.019)
Shock ω_t	1.184***	1.229***	1.211***	1.249***	1.279***	1.241***
	(0.146)	(0.138)	(0.105)	(0.111)	(0.101)	(0.091)
First stage ω_{t-1}	−3.901***	−3.628***	−3.824***	−3.546***	−3.113***	−3.226***
	(1.145)	(0.945)	(0.910)	(0.800)	(0.704)	(0.731)
First stage ω_t	−2.918*	−2.276*	−2.372*	−2.885***	−2.350***	−2.420***
	(1.647)	(1.294)	(1.279)	(0.967)	(0.815)	(0.819)
Panel B. Demand equation						
Demand elast. β_d	−0.028	−0.055**	−0.054**	−0.034	−0.062***	−0.066***
	(0.021)	(0.024)	(0.022)	(0.023)	(0.022)	(0.021)
First stage ω_t	−5.564***	−4.655***	−4.770***	−5.354***	−4.445***	−4.332***
	(1.489)	(1.300)	(1.249)	(1.384)	(1.210)	(1.186)

From Hughes, Knittel, Sperling (2008)

> As it turns out, identifying appropriate instrumental variables for gasoline demand is difficult. In this paper we experiment with crude oil production disruptions as instrumental variables.[10] Disruptions are represented for three countries, Venezuela (VZ_{jt}), Iraq (IQ_{jt}) and the United States (US_{jt}). These three countries were selected because each has had its production of crude oil affected by external shocks that are unlikely to be related to gasoline demand shocks. In Venezuela, a strike by oil workers beginning in December 2002 cut production to near zero and has significantly affected output for several years. In Iraq, an international embargo and more recently war have caused major disruptions to oil operations. In the United States, production has been in steady decline since the 1970s due to declining resources. In 2005, hurricanes Katrina and Rita resulted in the temporary loss of several hundred thousand barrels of production in the Gulf of Mexico.

Simultaneity more examples

- Download speeds and mobile data consumption: Consumers use their smart phones more when download speeds are fast, but high usage leads to congestion, which slows down speeds.
- Road investment and travel demand: better roads are more appealing to travelers, and highly-used routes are likely to receive more funding .
- Public opinion influences policy, and policy influences public opinion (cigarettes, parental leave, same-sex marriage).
- Intensity of policing and crime rates.

Measurement Error

The problem: the X variable is measured with noise

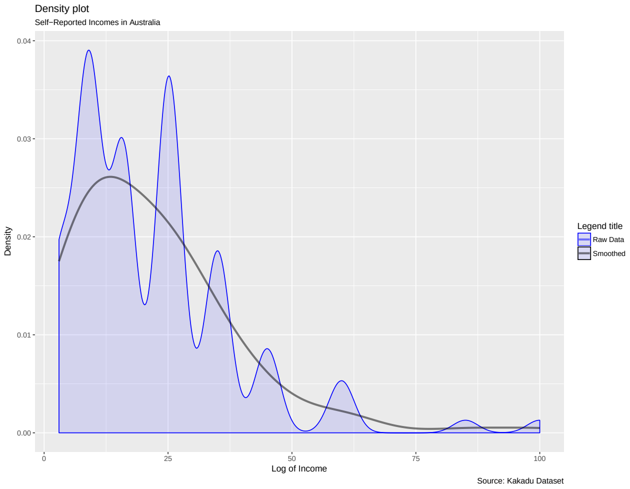
- The idea mathematically:

$$\textbf{TRUTH: } Y = \beta_0 + \beta_1 X + \varepsilon$$
$$\textbf{DATA: } Y = \beta_0 + \beta_1 X^* + \varepsilon^*$$

  where X^* is a noisy measure of X
- Examples:
  - Recording errors in data entry
  - Recollection errors in survey data (these are frequent)
  - Rounding errors
  - Hard to measure variables (e.g., a firm's capital stock)
- Note: we can also have measurement error in Y. Turns out these behave differently

Visualizing Measurement Error: Self-Reported Australian Incomes

Survey question: What is your income in thousands?



- Consider adding and subtracting noise to the truth:

$$\begin{aligned} Y_i =& \beta_0 + \beta_1 X_i + \varepsilon_i \\ =& \beta_0 + \beta_1 X_i^* + \underbrace{\beta_1 (X_i - X_i^*) + \varepsilon_i}_{\text{“New” Error Term:} V_i} \\ =& \beta_0 + \beta_1 X_i^* + V_i \end{aligned}$$

- X_i^* and V_i likely to be correlated because X^* is part of V
- Extra assumptions:
  - **Classical Errors-in-Variables:**

$$X_i = X_i^* + u_i n^{-1} \sum_{i=1}^{n} m\left(y_i, \mathbf{x}_i; \theta\right)$$

Under classical measurement error:

$$Y_i = \beta_0 + \beta_1 X_i^* + \underbrace{\beta_1 u_i + \varepsilon_i}_{V_i}$$

- Using the population definition of $\hat{\beta}^{OLS}$:

$$\begin{aligned}
\hat{\beta}^{OLS} \to & \frac{Cov(X^*, Y)}{Var(X^*)} \\
= & \frac{Cov(X + u, \beta_0 + \beta_1 X + \varepsilon)}{Var(X + u)} \\
= & \frac{Cov(X, \beta_0 + \beta_1 X + \varepsilon) + Cov(u, \beta_0 + \beta_1 X + \varepsilon)}{Var(X + u)} \\
= & \frac{\beta_1 Cov(X, X) + 0}{Var(X + u)}
\end{aligned}$$

- Continuing from above:

$$
\begin{aligned}
\hat{\beta}^{OLS} \to & \frac{\beta_1 Cov(X,X) + 0}{Var(X+u)} \\
= & \beta_1 \times \frac{Var(X)}{Var(X+u)} \\
= & \beta_1 \times \underbrace{\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2}}_{\text{Attenuation}}
\end{aligned}
$$

- Estimated coefficient converges to the truth times an attenuation term
  - Attenuation term is less than 1 $\Rightarrow$ pushes coefficient towards zero
    - **NB:** It does *not* make the term more negative—it dampens the coefficient and preserves the sign
  - This “attenuates” the effect of X on Y. Hence the name: **attenuation bias**

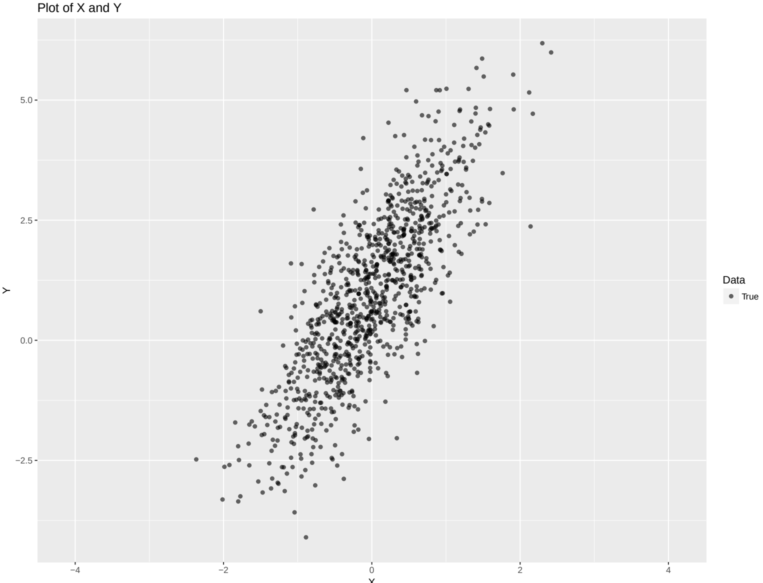
Measurement Error in X

- Why does measurement error attenuate the estimated effect of X on Y?
  - Noise in measured X makes it more likely to see high X and low X with some Y because of randomness
  - Extreme example: fix X and keep adding noise—eventually X^* will look like noise itself
- Notice the following rearranging of the attenuation term:

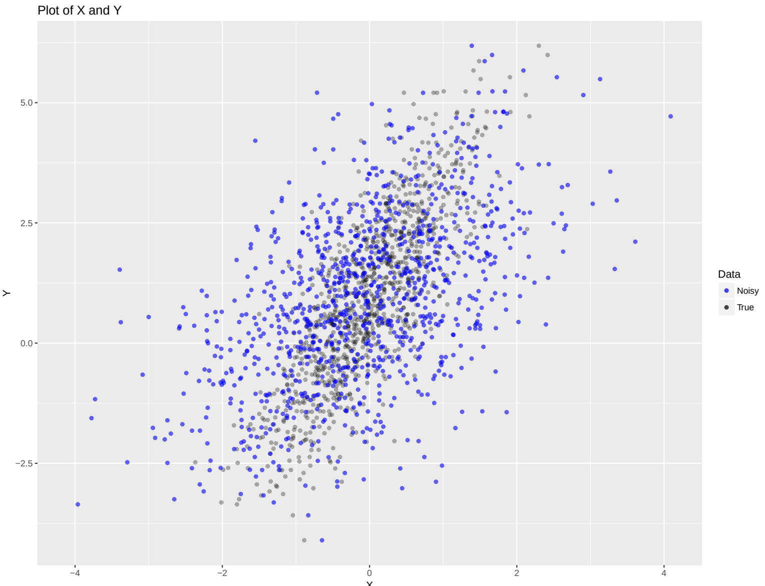
$$\frac{\sigma_X^2}{\sigma_X^2 + \sigma_u^2} = \frac{1}{1 + (\sigma_u^2/\sigma_X^2)}$$

  - σ_X^2/σ_u^2 is called the **signal-to-noise ratio**
  - Larger signal to noise ratio $\Rightarrow$ smaller bias
  - What matter is the *relative* variance of X to u
  - If σ_u^2 is small *relative* to σ_X^2 then attenuation bias isn't too large

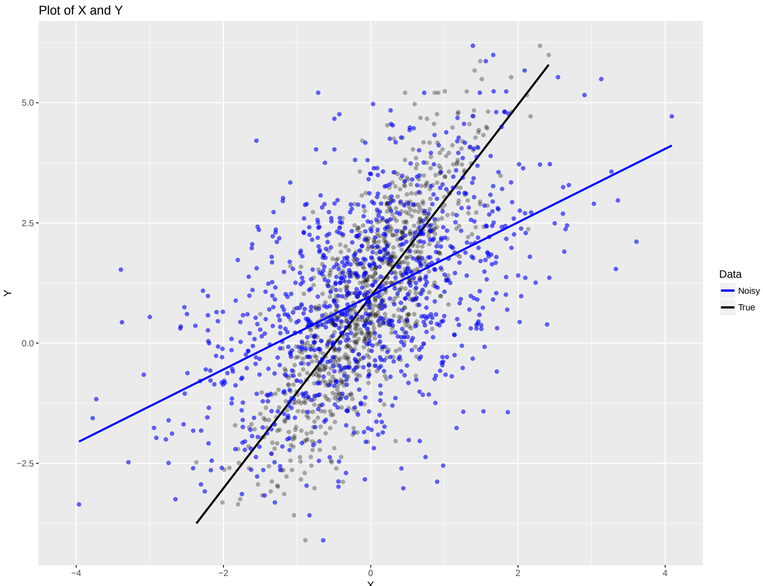
Visualizing Measurement Error



Visualizing Measurement Error



Visualizing Measurement Error



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- How do we address attenuation bias using instrumental variables?

Measurement Error in Y

- Suppose the econometrician observes $Y^* = Y + u$, where Y is the true value.

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\ Y_i + Y_i^* &= \beta_0 + \beta_1 X_i + \varepsilon_i + Y_i^* \\ Y_i^* &= \beta_0 + \beta_1 X_i + \varepsilon_i + (Y_i^* - Y_i) \end{aligned}$$

- Again, we can see this as changing the error term:

$$Y_i^* = \beta_0 + \beta_1 X_i + V_i$$

  where $V_i = \varepsilon_i + (Y_i^* - Y_i)$.

- As long as this is classical measurement error, i.e. $\mathbb{E}[u \mid X] = 0$, then as long as exogeneity is satisfied with the original error term $\mathbb{E}[\varepsilon \mid X] = 0$, it will still be satisfied for the modified error term $\mathbb{E}[V \mid X] = 0$.